

Article

An Energy Saving MTPA-Based Model Predictive Control Strategy for PMSM in Electric Vehicles Under Variable Load Conditions

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Abstract

To promote energy efficiency and support sustainable electric transportation, this study addresses the challenge of real-time and energy-optimal control of permanent magnet synchronous motors (PMSMs) in electric vehicles operating under variable load conditions, proposing a novel Laguerre-based model predictive control (MPC) strategy integrated with maximum torque per ampere (MTPA) operation. Traditional MPC methods often suffer from limited prediction horizons and high computational burden when handling strong coupling and time-varying loads, compromising real-time performance. To overcome these limitations, a Laguerre function approximation is employed to model the dynamic evolution of control increments using a set of orthogonal basis functions, effectively reducing the control dimensionality while accelerating convergence. Furthermore, to enhance energy efficiency, the MTPA strategy is embedded by reformulating the current allocation process using d- and q-axis current variables and deriving equivalent reference currents to simplify the optimization structure. A cost function is designed to simultaneously ensure current accuracy and achieve maximum torque per unit current. Simulation results under typical electric vehicle conditions demonstrate that the proposed Laguerre-MTPA MPC controller significantly improves steady-state performance, reduces energy consumption, and ensures faster response to load disturbances compared to traditional MTPA-based control schemes. This work provides a practical and scalable control framework for energy-saving applications in sustainable electric transportation systems.

Keywords: energy efficiency; electric vehicles; model predictive control; maximum torque per ampere



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1. Introduction

In electric vehicle applications, which are increasingly regarded as a cornerstone of energy-saving and sustainable transportation, the energy efficiency and thermal reliability of permanent magnet synchronous motors (PMSMs) largely depend on the proper coordination of d-axis and q-axis stator currents. When the current distribution deviates from the optimal condition, such as when the d-axis and q-axis components are not effectively coordinated, the motor must increase the total current amplitude to sustain the required torque output [1–3]. This non-optimal operation results in higher copper losses, increased energy consumption, and accelerated temperature rise, thereby imposing greater

thermal stress on the system. Over time, such conditions can significantly compromise the long-term reliability and operational lifespan of PMSMs. As electric vehicles continue to evolve toward greener and more efficient mobility solutions, optimizing current allocation has become a critical aspect in the development of energy-efficient control strategies for sustainable electric transportation systems.

Achieving efficient current allocation under dynamic load conditions is a key challenge for improving the overall energy efficiency and operational stability of PMSM drive systems. This issue becomes particularly critical in electric vehicle scenarios, where the motor is frequently subjected to high-load, nonlinear disturbances and rapidly shifting operating points. In such environments, it is essential for the motor to maintain the required torque output while minimizing current consumption to prevent unnecessary energy losses and thermal runaway. As a result, energy-optimized control strategies that aim to minimize current amplitude without compromising torque performance have emerged as a vital direction in the evolution of advanced motor control technologies [4,5].

The Maximum Torque Per Ampere (MTPA) [6–8] control strategy has been widely adopted as a classical and effective method for the energy-efficient control of PMSMs under steady-state or quasi-steady-state conditions. Its core principle lies in precisely adjusting the ratio between the d-axis and q-axis currents to achieve the maximum electromagnetic torque output per unit of stator current, thereby minimizing copper losses and improving system efficiency. This strategy is particularly effective for PMSMs with saliency, where magnetic flux linkage coupling exists between the axes. Studies [9] have shown that compared to traditional vector control methods based solely on q-axis orientation, MTPA control can significantly reduce current consumption, suppress thermal buildup, and improve operational stability, especially in applications characterized by stringent current constraints or prolonged high-load conditions.

In practical engineering applications, the most commonly implemented MTPA method is based on a dual closed-loop PI vector control structure, which includes an outer speed loop and an inner current loop. The speed controller first generates a desired torque command, and then, using the MTPA characteristic curve, the optimal reference values of d-axis and q-axis currents are computed. These reference currents are regulated via PI controllers to generate the appropriate control voltages for the inverter. Owing to its clear structure and well-established implementation, this approach has been widely applied in industrial servos, CNC machine tools, and electric vehicle drives [10–13]. However, the PI-based implementation suffers from critical limitations under varying operating conditions, primarily due to its high sensitivity to motor parameter variations. As the motor parameters shift with temperature, load, or prolonged operation, the fixed gain settings of the PI controller may become suboptimal, leading to degraded dynamic performance. Furthermore, the accuracy of MTPA reference current calculation heavily depends on motor parameters such as flux linkage and inductance. Modeling inaccuracies and parameter drifts can cause deviations from the optimal current trajectory, resulting in torque fluctuations and reduced energy efficiency [14–16].

To further enhance dynamic response and control robustness, model predictive control (MPC) has been increasingly integrated into MTPA strategies in recent years as a promising alternative to traditional PI-based control structures [17–21]. MPC operates by predicting the system's future behavior over a finite time horizon based on current state measurements and solving an optimization problem at each control interval. This rolling optimization explicitly accounts for system constraints and multiple performance objectives, enabling more accurate and adaptive control. Depending on whether the switching states of the inverter are explicitly considered, MPC methods are generally categorized into finite control set MPC (FCS-MPC) [22–24] and continuous control set MPC (CCS-MPC) [25]. FCS-MPC

is widely used in motor control applications due to its simplicity and high computational efficiency. It directly searches for the optimal switching state within a finite set, eliminating the need for modulation. However, the discrete nature of the control input leads to coarse control granularity, which may result in steady-state errors and limited robustness to parameter variations. In contrast, CCS-MPC computes a continuous-valued control voltage that must be applied through a modulation scheme, offering smoother control signals and superior robustness to system uncertainties. Nevertheless, CCS-MPC typically incurs a significantly higher computational burden, posing challenges for real-time implementation in high-frequency motor drive systems.

In existing studies [26], MTPA reference current trajectories are typically derived through analytical expressions or numerical approximation methods and incorporated into the MPC cost function as part of the control optimization objective. One representative FCS-MPC approach introduces the MTPA current error into a single-objective cost function, selecting control inputs that guide the current trajectory as close as possible to the optimal MTPA path [27]. To improve control accuracy and system consistency, subsequent research has formulated multi-objective cost functions that combine torque error, MTPA tracking error, and current constraint penalties [28]. However, due to the inherent nonlinearity of MTPA characteristics, the resulting cost functions often involve high-order terms, leading to increased computational complexity and posing challenges for real-time optimization. Alternative approaches have attempted to incorporate absolute or squared trajectory deviation terms into the cost function, sometimes combined with dynamic weighting mechanisms to enhance control smoothness. Yet, these formulations often suffer from structural inconsistency and convergence instability, which limit the practical deployment of MTPA-MPC strategies in industrial motor drives [29]. Furthermore, while FCS-MPC generally lacks the adaptive capability to cope with motor parameter variations, resulting in steady-state errors, CCS-MPC offers better robustness due to its embedded integral action. Nevertheless, CCS-MPC remains constrained in real-world applications by its high computational load and strong dependence on hardware performance. As such, recent research efforts are increasingly focused on developing MTPA-MPC schemes that are lightweight in structure, analytically tractable in cost function formulation, and capable of dynamically adjusting current trajectories. The ultimate goal is to achieve accurate MTPA current tracking and optimal control performance while maintaining low computational overhead suitable for real-time electric drive applications.

In this study, a low-dimensional, high-response Laguerre-based model predictive control algorithm is developed within the MPC framework to address the computational challenges of real-time motor control. By parameterizing the control input sequence using Laguerre functions, the proposed method significantly reduces the number of optimization variables, thereby lowering the computational burden and improving the online solving efficiency of the controller. This dimensionality reduction enables the practical implementation of continuous control set MPC in high-frequency applications such as electric vehicle drives. Building upon this foundation, the proposed strategy further integrates the MTPA control principle to enhance current utilization and torque density. Specifically, the d-axis and q-axis currents are selected as the primary predictive state variables, and an equivalent current construction method is introduced to simplify the generation of MTPA reference currents. A current allocation cost function is designed to balance tracking accuracy and energy efficiency, enabling the motor to produce maximum torque per unit current. As a result, the proposed Laguerre-MTPA MPC approach achieves energy-efficient, high-performance torque control under dynamic load conditions, offering a promising solution for sustainable and real-time electric transportation systems.

2. The PMSM Model and MTPA-Based Model Predictive Control

2.1. PMSM Model

The continuous-time electrical dynamics equations and mechanical equation of a PMSM in the dq0 rotating reference frame can be described by [30]:

$$\begin{aligned}\frac{di_d}{dt} &= -\frac{R_s}{L_d}i_d + \frac{L_q}{L_d}P\omega_m i_q + \frac{u_d}{L_d} \\ \frac{di_q}{dt} &= -\frac{R_s}{L_q}i_q + \frac{L_d}{L_q}P\omega_m i_d - \frac{\phi_f}{L_q}P\omega_m + \frac{u_q}{L_q}\end{aligned}\quad (1)$$

$$T_e = \frac{3}{2}P[\phi_f i_q + (L_d - L_q)i_d i_q] \quad (2)$$

where i_d is the stator current of the d -axis; i_q is the stator current of the q -axis; ω_m is the mechanical angular speed of the rotor; ϕ_f is the base electromotive force constant; R_s is the stator winding resistance; L_d is the stator inductance of the d -axis; L_q is the stator inductance component of the q -axis; P is the number of pole pairs; u_d is the stator voltage of the d -axis; u_q is the stator voltage of the q -axis; and T_e is the electromagnetic torque.

Assuming the sampling interval is T_s , the items $\omega_m i_q(k)$ and $\omega_m i_d(k)$ are considered as constant variables in a sampling interval; then, (1) can be discretized as:

$$\begin{aligned}i_d(k+1) &= \frac{L_d - T_s R_s}{L_d} i_d(k) + \frac{L_q}{L_d} T_s P \omega_m i_q(k) + \frac{T_s}{L_d} u_d(k) \\ i_q(k+1) &= \frac{L_q - T_s R_s}{L_q} i_q(k) - \frac{L_d}{L_q} T_s P \omega_m i_d(k) - T_s P \frac{\phi_f}{L_q} \omega_m(k) + \frac{T_s}{L_q} u_q(k)\end{aligned}\quad (3)$$

Assume the d -axis current i_d , q -axis current i_q , mechanical angular speed ω_m , the product of i_d and ω_m , and the product of i_q and ω_m are the state variables. Assuming the d -axis voltage u_d and q -axis voltage u_q are the input variables and d -axis current i_d and q -axis current i_q are the output variables, then (3) can be described in the state space model:

$$\begin{aligned}x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k)\end{aligned}\quad (4)$$

$$x(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \\ \omega_m(k) \\ \omega_m i_d(k) \\ \omega_m i_q(k) \end{bmatrix}, y(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}, u(k) = \begin{bmatrix} u_d(k) \\ u_q(k) \end{bmatrix}, \quad (5)$$

$$B = \begin{bmatrix} \frac{T_s}{L_d} & 0 & 0 & 0 & 0 \\ 0 & \frac{T_s}{L_q} & 0 & 0 & 0 \end{bmatrix}^T, C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}^2$$

$$A = \begin{bmatrix} \frac{L_d - T_s R_s}{L_d} & 0 & 0 & 0 & \frac{T_s P L_q}{L_d} \\ 0 & \frac{L_q - T_s R_s}{L_q} & -\frac{T_s P L_d}{L_q} & 0 & -\frac{T_s P \phi_f}{L_q} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (6)$$

2.2. MTPA Strategy

From (2), it can be seen that the torque of a PMSM consists of two parts: the first part is the electromagnetic torque, and the second part is the reluctance torque. When using the $i_d = 0$ control strategy, only electromagnetic torque takes effect, and the torque is determined by the q -axis current. However, this control strategy will reduce the motor

operating efficiency and the torque performance. In order to better utilize reluctance torque, the control strategy of MTPA can be adopted to control the motor.

The MTPA control method refers to the minimum stator current of the motor under the same electromagnetic torque, which means (2) is subject to [31]:

$$i_s = \min \sqrt{i_d^2 + i_q^2} \quad (7)$$

In order to obtain the d -axis and q -axis currents of the PMSM under the MTPA control method, the following Lagrange function is usually constructed:

$$F = \sqrt{i_d^2 + i_q^2} + \lambda \left(T_e - \frac{3}{2} P (\phi_f i_q + (L_d - L_q) i_d i_q) \right) \quad (8)$$

Considering $i_d \leq 0$, $i_q \geq 0$, note that the constraints $i_d \leq 0$ and $i_q \geq 0$ are introduced only for the forward-driving operation of the traction system. A negative i_d is a common requirement in MTPA control of IPMSMs to utilize the reluctance torque, and it does not lead to demagnetization as long as the current remains within the safe operating range defined by the machine design. The constraint $i_q \geq 0$ corresponds to forward torque generation. If motor reversal is required, the condition can be relaxed to $i_q < 0$, which produces negative torque. In this paper, only the forward-driving case is considered for clarity.

λ is the Lagrange factor.

Then, take the partial derivative of function F with respect to the variables i_d , i_q , and λ .

$$\begin{aligned} \frac{\partial F}{\partial i_d} &= \frac{i_d}{\sqrt{i_d^2 + i_q^2}} - \frac{3}{2} \lambda P (L_d - L_q) i_q = 0 \\ \frac{\partial F}{\partial i_q} &= \frac{i_q}{\sqrt{i_d^2 + i_q^2}} - \frac{3}{2} \lambda P \phi_f - \frac{3}{2} \lambda P (L_d - L_q) i_d = 0 \\ \frac{\partial F}{\partial \lambda} &= T_e - \frac{3}{2} P [\phi_f i_q + (L_d - L_q) i_d i_q] = 0 \end{aligned} \quad (9)$$

By solving the equations, the relationship between the d -axis and q -axis currents under the MTPA control method can be described as [32]:

$$\left(i_d - \frac{\phi_f}{2(L_q - L_d)} \right)^2 = \left(\frac{\phi_f}{2(L_q - L_d)} \right)^2 + i_q^2 \quad (10)$$

considering that $i_d \leq 0$, $i_q \geq 0$.

From (7) and (10), it can be seen that the relationship between d -axis and q -axis currents is not linear and is difficult to solve. Therefore, a simpler method for solving d -axis and q -axis currents is needed under the MTPA control method.

2.3. Traditional MTPA Model Predictive Control

Based on the mathematical model of PMSM described in (4), the MPC of PMSM usually calculates the system behavior under different control sequences within a finite predictive period. A preset cost function is used to obtain the optimal control sequence; then, the first control variable in the optimal control sequence is applied on the PMSM control to achieve rolling optimization.

The traditional MTPA-based MPC algorithm first calculates the MTPA reference currents through (7) and (10) and then obtains the optimal voltage by solving the cost function.

The cost function usually includes current errors, or MTPA curve errors, or a combination of both described in (10) [33,34].

$$J = \sum_{m=1}^{N_P} \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \end{bmatrix} \right)^T q_i \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \end{bmatrix} \right) + q_{mtpa} \sum_{m=1}^{N_P} \left| i_d^2(k+m) + \frac{\phi_f}{(L_d - L_q)} i_d(k+m) - i_q^2(k+m) \right| \quad (11)$$

where i_d^* and i_q^* are the references of the d -axis current and q -axis current, respectively; m is the future m -th predictive period starting from k ; n is the future n -th control period starting from k ; q_i is the weight of current errors; and q_{mtpa} is the weight of MTPA curve errors.

From (11), it can be seen that the second item is an absolute expression, which needs to be segmented according to the situation when solving the optimal control variable. Moreover, it differs from the standard quadratic programming cost function and cannot be directly solved using the quadratic programming solutions, which increase the difficulty of the solution. Therefore, a more convenient cost function is needed for MTPA-based MPC control for PMSM.

3. The Proposed MTPA-Based MPC Method

3.1. The Proposed MTPA Method

At every sampling period k , the d -axis and q -axis currents of PMSM can be represented by a unique variable i_N :

$$i_N(k) = i_q(k) - i_d(k) \quad (12)$$

$$\text{Considering that } i_N(k) = \begin{cases} (i_q(k) - i_d(k)) > 0 & i_q(k) > 0 \\ 0 & i_q(k) = 0 \end{cases}$$

First, (10) can be transformed into:

$$(i_d(k) - \Lambda + i_q(k))(i_d(k) - \Lambda - i_q(k)) = \Lambda^2 \quad (13)$$

where $\Lambda = \frac{\phi_f}{2(L_q - L_d)}$.

Then, substituting (12) into (10), we obtain:

$$-(i_d(k) - \Lambda + i_q(k))(i_N(k) + \Lambda) = \Lambda^2 \quad (14)$$

Next, combining (12) and (14), an equation system is constructed as follow:

$$\begin{cases} i_d(k) + i_q(k) = \Lambda - \frac{\Lambda^2}{i_N(k) + \Lambda} \\ i_q(k) - i_d(k) = i_N(k) \end{cases} \quad (15)$$

By solving (15), we can obtain:

$$\begin{cases} i_q^*(k) = \frac{i_N^2(k) + 2\Lambda i_N(k)}{2(i_N(k) + \Lambda)} \\ i_d^*(k) = -\frac{i_N^2(k)}{2(i_N(k) + \Lambda)} \end{cases} \quad (16)$$

The $i_d^*(k)$ and $i_q^*(k)$ in (16) is the reference current under the MTPA control method. Compared with (7) and (10), (16) is used to calculate the current of the d -axis and q -axis, which reduces the load of calculation. Figure 1 shows the schematic diagram of the proposed MTPA algorithm.

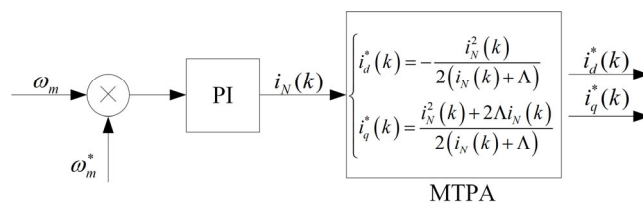


Figure 1. The proposed MTPA schematic diagram.

3.2. The Proposed Cost Function

There is a segmented function in (11), which is complex to calculate. Then, a new cost function is designed according to the MTPA characteristics as follows:

$$J = \sum_{m=1}^{N_p} \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^*(k+m) \\ i_q^*(k+m) \end{bmatrix} \right)^T q_i \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^*(k+m) \\ i_q^*(k+m) \end{bmatrix} \right) + q_{mtpa} \sum_{m=1}^{N_p} \left(i_d(k+m) - \Lambda \right)^2 - \Lambda^2 - i_q^2(k+m) \Big)^2 + \sum_{n=1}^{N_c} \Delta u^T(k+n-1) r \Delta u(k+n-1) \quad (17)$$

The cost function in (17) consists of three parts: total current errors within N_p predictive periods, MTPA curve errors within N_p predictive periods, and voltage increment errors within N_c control periods. The total current errors refers to the errors between the operating current of the PMSM and the reference currents of the PMSM under MTPA control. The purpose of this item is to make the currents of the PMSM follow the reference currents as much as possible, that is, the reference currents of the PMSM under MTPA control. The second MTPA curve errors is to ensure that the d-axis and q-axis current trajectories of the PMSM are as close as possible to the MTPA curve during operation, further ensuring that the PMSM operates in MTPA mode. When the PMSM becomes stable, the voltage increments should approach zero as soon as possible to ensure the steady-state performance and dynamic response of the PMSM.

In the second item of (17), the power of expression J is 4. To simplify calculations, the power of the cost function must be reduced.

First, the second item in (17) can be reconstructed:

$$\begin{aligned} & (i_d(k+m) - \Lambda)^2 - \Lambda^2 - i_q^2(k+m) \\ &= (i_d(k+m) - \Lambda - i_q(k+m))(i_d(k+m) - \Lambda + i_q(k+m)) - \Lambda^2 \end{aligned} \quad (18)$$

Substitute (12) into (17), then:

$$\begin{aligned} & (i_d(k+m) - \Lambda)^2 - \Lambda^2 - i_q^2(k+m) \\ &= -(i_N(k+m) + \Lambda)(i_d(k+m) + i_q(k+m) - \Lambda) - \Lambda^2 \end{aligned} \quad (19)$$

When a PMSM is running, the current should be kept as close to the MTPA curve as possible. Therefore, the following equation is true:

$$\begin{aligned} & -(i_N(k+m) + \Lambda)(i_d(k+m) + i_q(k+m) - \Lambda) - \Lambda^2 = 0 \\ & i_d(k+m) + i_q(k+m) = \frac{\Lambda i_N(k+m)}{\Lambda + i_N(k+m)} \end{aligned} \quad (20)$$

According to the analysis above, the cost function (16) is equal to:

$$J = \sum_{m=1}^{N_p} \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \end{bmatrix} \right)^T q_i \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \end{bmatrix} \right) + q_{mtpa} \sum_{m=1}^{N_p} \left(i_d(k+m) + i_q(k+m) - \frac{\Lambda i_N(k+m)}{\Lambda + i_N(k+m)} \right)^2 + \sum_{n=1}^{N_c} \Delta u^T(k+n-1) r \Delta u(k+n-1) \quad (21)$$

Furthermore, the cost function (21) can be changed to:

$$J = \sum_{m=1}^{N_p} \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \\ i_d(k+m) + i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \\ i_{mtpa}^* \end{bmatrix} \right)^T q \left(\begin{bmatrix} i_d(k+m) \\ i_q(k+m) \\ i_d(k+m) + i_q(k+m) \end{bmatrix} - \begin{bmatrix} i_d^* \\ i_q^* \\ i_{mtpa}^* \end{bmatrix} \right) + \sum_{n=1}^{N_c} \Delta u^T(k+n-1) r \Delta u(k+n-1) \quad (22)$$

$$\text{where } q = \begin{bmatrix} q_{id} & 0 & 0 \\ 0 & q_{iq} & 0 \\ 0 & 0 & q_{mtpa} \end{bmatrix}, i_{mtpa}^* = \frac{\Lambda i_N(k+m)}{\Lambda + i_N(k+m)}.$$

Assume the following:

$$y_m(k+m) = \begin{bmatrix} i_d(k+m) \\ i_q(k+m) \\ i_d(k+m) + i_q(k+m) \end{bmatrix} = \begin{bmatrix} i_d(k+m) \\ i_q(k+m) \\ i_{mtpa}(k+m) \end{bmatrix} \quad (23)$$

$$y_m^*(k+m) = \begin{bmatrix} i_d^*(k+m) \\ i_q^*(k+m) \\ i_{mtpa}^*(k+m) \end{bmatrix} = \begin{bmatrix} i_N^2(k+m) + 2\Lambda i_N(k+m) \\ 2(i_N(k+m) + \Lambda) \\ -\frac{i_N^2(k+m)}{2(i_N(k+m) + \Lambda)} \\ \frac{\Lambda i_N(k+m)}{\Lambda + i_N(k+m)} \end{bmatrix}$$

Then, substituting (23) into (22), we can obtain:

$$J = \sum_{m=1}^{N_p} (y_m(k+m) - y_m^*(k+m))^T q (y_m(k+m) - y_m^*(k+m)) + \sum_{n=1}^{N_c} \Delta u^T(k+n-1) r \Delta u(k+n-1) \quad (24)$$

Equation (24) is the proposed cost function for solving the optimal control variables.

3.3. The Proposed PMSM Model

From (22) and (23), it can be seen that the sum of the d-axis current and q-axis current has been added to the output variables and cost function. In order to find the solution of the optimal variables, it is necessary to modify the state space model of PMSM described in (4).

First, modify the state space model of PMSM described in (4) to the 1st state space model, namely the incremental model.

$$\Delta x(k+1) = A_1 \Delta x(k) + B_1 \Delta u(k) \quad (25)$$

$$y(k) = C_1 \Delta x(k)$$

$$\text{where } \Delta x(k) = \begin{bmatrix} \Delta i_d(k) \\ \Delta i_q(k) \\ \Delta \omega_m(k) \\ \Delta \omega_m i_d(k) \\ \Delta \omega_m i_q(k) \\ i_d(k) \\ i_q(k) \end{bmatrix}, \quad y(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \end{bmatrix}, \quad \Delta u(k) = \begin{bmatrix} \Delta u_d(k) \\ \Delta u_q(k) \end{bmatrix},$$

$$B_1 = \begin{bmatrix} \frac{T_s}{L_d} & 0 & 0 & 0 & 0 & \frac{T_s}{L_d} & 0 \\ 0 & \frac{T_s}{L_q} & 0 & 0 & 0 & 0 & \frac{T_s}{L_q} \end{bmatrix}^T, \quad C_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_1 = \begin{bmatrix} \frac{L_d - T_s R_s}{L_d} & 0 & 0 & 0 & \frac{T_s P L_q}{L_d} & 0 & 0 \\ 0 & \frac{L_q - T_s R_s}{L_q} & -\frac{T_s P L_d}{L_q} & 0 & -\frac{T_s P \phi_f}{L_q} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{L_d - T_s R_s}{L_d} & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \frac{L_q - T_s R_s}{L_q} & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

Next, add the sum of the d -axis current and q -axis current to the 1st state space model (25), and the 2nd state space model is formed in (25).

$$\begin{aligned} x_m(k+1) &= A_m x_m(k) + B_m \Delta u(k) \\ y_m(k) &= C_m x_m(k) \end{aligned} \quad (26)$$

$$\text{where } x_m(k) = \begin{bmatrix} \Delta i_d(k) \\ \Delta i_q(k) \\ \Delta \omega(k) \\ \Delta i_d \omega(k) \\ \Delta i_q \omega(k) \\ i_d(k) \\ i_q(k) \\ i_d(k) + i_q(k) \end{bmatrix}, \quad y_m(k) = \begin{bmatrix} i_d(k) \\ i_q(k) \\ i_d(k) + i_q(k) \end{bmatrix}, \quad \Delta u(k) = \begin{bmatrix} \Delta u_d(k) \\ \Delta u_q(k) \end{bmatrix},$$

$$A_m = \begin{bmatrix} A_1 & 0 & 0 \\ C_1 A_1 & 1 & 0 \\ C_z C_1 A_1 & 0 & 1 \end{bmatrix}, \quad B_m = \begin{bmatrix} B_1 \\ C_1 B_1 \\ C_z C_1 B_1 \end{bmatrix}, \quad C_m = [0_{3 \times 5} \quad I_{3 \times 3}], \quad C_z = [1 \quad 1].$$

3.4. The Improved PMSM Model Using the LAGUERRE Function

A stable linear system can be represented by a series of Laguerre functions. The differences of voltage $\Delta u(k)$ described in (26) will be zero when the PMSM becomes steady; in order to accelerate the rate of voltage increment decay to zero, a Laguerre function is introduced to represent the voltage increment. By adjusting the parameters of the Laguerre function, we can reduce the number of predictive periods and speed up the system response.

$$\Delta u(k) = \begin{bmatrix} \Delta u_d(k) \\ \Delta u_q(k) \end{bmatrix} = \begin{bmatrix} L_d(0) \eta_d \\ L_q(0) \eta_q \end{bmatrix} = L_u(0)^T \eta_I \quad (27)$$

$$\text{where } \eta_I = \begin{bmatrix} \eta_d \\ \eta_q \end{bmatrix} = \begin{bmatrix} c_1^d & \cdots & c_{N_d}^d & c_1^q & \cdots & c_{N_q}^q \end{bmatrix}^T,$$

$$L_u(0) = \begin{bmatrix} L_d(0) \\ L_q(0) \end{bmatrix} = \begin{bmatrix} l_1^d(0) & \cdots & l_{N_d}^d(0) & 0 & 0 & 0 \\ 0 & 0 & 0 & l_1^q(0) & \cdots & l_{N_q}^q(0) \end{bmatrix}.$$

η_l is the Laguerre coefficients matrix. $c_1^d, c_2^d \dots c_{N_d}^d$ are the Laguerre coefficients of the d-axis voltage; $c_1^q, c_2^q \dots c_{N_q}^q$ are the Laguerre coefficients of the q-axis. N_d is the number of items of the d-axis voltage Laguerre functions. N_q is the number of items of the q-axis voltage Laguerre functions. $L_u(0)$ is the Laguerre vector of the d-axis voltage and q-axis voltage.

To ensure analytical clarity, the assumptions used in transforming the cost function are as follows: (i) the MTPA deviation term is replaced by a quadratic surrogate to guarantee differentiability and compatibility with quadratic programming; (ii) the sum of i_d and i_q is introduced into the outputs so that the MTPA manifold can be softly enforced within the cost; and (iii) an incremental model with Laguerre parameterization is employed to represent input increments, which preserves convexity while reducing dimensionality. Under these assumptions, the final cost function can be written as a quadratic form in the optimization vector, with a positive semidefinite Hessian. When the increment weight $r_u > 0$ and the Laguerre embedding has full column rank, the Hessian is strictly positive definite, ensuring convexity and uniqueness of the solution. For closed-loop stability, the incremental MPC framework yields an ISS-type Lyapunov decrease condition, since the smoothing term $\sum r_u \|\Delta u\|^2$ guarantees the boundedness of increments and the Laguerre factor $a \in (0, 1)$ ensures exponential decay. Finally, the tuning criteria of the weights are summarized as follows: q_d, q_q balance d- and q-axis current tracking; r_{mtpa} determines the adherence to the analytical MTPA curve; and r_u provides smoothness and stability margin. A sensitivity test shows that varying each weight by one order of magnitude changes current RMS, overshoot, and settling time in a predictable monotonic fashion, which confirms robustness of the proposed settings without additional graphical evidence.

The cost function weights have distinct physical roles: q_d and q_q penalize tracking errors of the d- and q-axis currents, q_{mtpa} enforces proximity to the analytical MTPA trajectory, and r penalizes voltage increments to smooth the control action and improve closed-loop stability. In practice, the weights are tuned empirically: larger q_d and q_q accelerate current tracking at the expense of overshoot, larger q_{mtpa} enhance efficiency but slow down the transient, and larger r reduce torque/current ripple but lead to slower dynamic response. In the following simulation parameter settings, the nominal values are used ($q_d = q_q = q_{MTPA} = 10^4, r = 0.1$), which were found to balance accuracy, efficiency, and robustness. Sensitivity analysis shows that varying each weight by one order of magnitude results in monotonic and predictable changes in RMS current, overshoot, and settling time, confirming the robustness of the tuning strategy.

3.5. The Constrains of Currents

It is well known that the currents of PMSM must satisfy

$$i_d^2 + i_q^2 \leq I_{max}^2 \quad (28)$$

where i_d, i_q are the d-/q-axis stator currents obtained from the Clarke–Park transformation, and I_{max} is the maximum admissible current amplitude. Since (28) is nonlinear, it must be linearized before embedding into the quadratic programming framework.

Combining (10), (13), and (28), we obtain:

$$\left(i_d - \frac{\Lambda}{2}\right)^2 \leq \frac{I_{max}^2}{2} + \frac{a^2}{4} \quad (29)$$

where $a = \Lambda$, and Λ is in (13). In addition, a common approach is to apply the $\lVert \cdot \rVert_1$ -norm inner approximation, $|i_d| + |i_q| \leq I_{max}$, which guarantees that if it holds, then (28) also holds. Expanding the absolute values gives four linear inequalities:

$i_d + i_q \leq I_{\max}, i_d - i_q \leq I_{\max}, -i_d + i_q \leq I_{\max}, -i_d - i_q \leq I_{\max}$, or in compact form, $S_4 \mathbf{i} \leq I_{\max} \mathbf{1}_4$, where $\mathbf{I}_{\text{stack}} = [\mathbf{i}(k+1)^\top, \dots, \mathbf{i}(k+N_p)^\top]^\top$. This procedure results in a conservative inner approximation; the feasible set changes from a circular disk of radius $I_{\max}$ (nonlinear) to an inscribed diamond (linear). Although this reduces the feasible region slightly, it ensures that the true current limit is never violated.

By solving (29), we can obtain:

$$\begin{cases} i_d \leq \frac{\Lambda}{2} + \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} \\ -i_d \leq \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} \end{cases} \quad (30)$$

According to (12), we can obtain:

$$\begin{cases} i_q \leq \frac{\Lambda}{2} + \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} + i_N \\ -i_q \leq \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} - i_N \end{cases} \quad (31)$$

Then, combining (10), (30), and (31), the constraints of the d -axis and q -axis current can be described as follows:

$$\begin{cases} i_d \leq 0 \\ -i_d \leq \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} \\ i_q \leq \frac{\Lambda}{2} + \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} + i_N \\ -i_q \leq \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} - i_N \end{cases} \quad \text{subject to: } \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} - i_N \leq 0 \quad (32)$$

or

$$\begin{cases} i_d \leq 0 \\ -i_d \leq \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} \\ i_q \leq \frac{\Lambda}{2} + \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} + i_N \\ -i_q \leq 0 \end{cases} \quad \text{subject to: } \sqrt{\frac{I_{\max}^2}{2} + \frac{a^2}{4}} - \frac{\Lambda}{2} - i_N > 0 \quad (33)$$

Based on the 2nd space state model (26), cost function (24) and (27), constraints (32) and (33), the incremental optimal control variable Δu can be obtained. The proposed MTPA-based MPC method is presented in Figure 2.

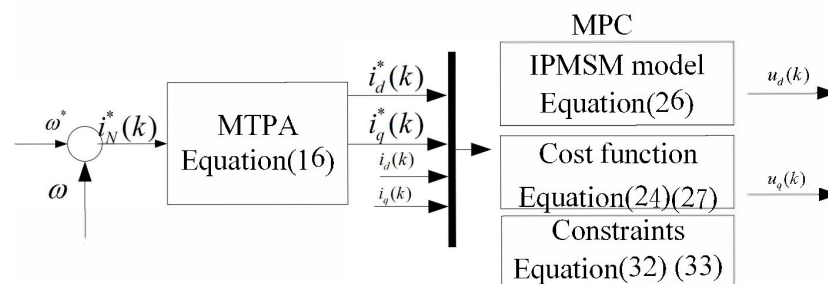


Figure 2. The proposed MTPA-based MPC schematic diagram.

4. Results

To validate the effectiveness of the proposed Laguerre-MTPA MPC, extensive simulation experiments were conducted using MATLAB/Simulink R2017a. Two types of PMSMs were selected to evaluate the performance of the proposed method under different saliency conditions. The motor parameters were designed according to Tables 1–3. Motor 1 exhibits a large difference between the d -axis and q -axis inductances, representing a high-saliency machine, while Motor 2 features a relatively small inductance difference, rep-

representing a low-saliency configuration. This setup allows for a comprehensive assessment of Laguerre-MTPA MPC adaptability and performance across various motor types. The listed weights were chosen to balance tracking accuracy, efficiency, and smoothness, as discussed in Section 3.4.

Table 1. Motor 1 parameters.

Parameter	Description	Value
P_n	Pole pair number	4
R_s	Stator winding resistance	0.958 Ω
L_d	Inductance along d-axis	0.00525 mH
L_q	Inductance along q-axis	0.012 mH
ψ_f	Permanent magnet flux linkage	0.1827 Wb
J	Moment of inertia	0.003 $\text{kg} \cdot \text{m}^2$
B	Viscous friction coefficient	0.0038 $\text{N} \cdot \text{m} \cdot \text{s}$

Table 2. Motor 2 parameters.

Parameter	Description	Value
P_n	Pole pair number	6
R_s	Stator winding resistance	0.00423 Ω
L_d	Inductance along d-axis	0.171 mH
L_q	Inductance along q-axis	0.391 mH
ψ_f	Permanent magnet flux linkage	0.1039 Wb

Table 3. Laguerre-MTPA MPC controller parameters.

Parameter	Description	Value
$k_{p\omega}$	Speed proportional gain	5
$k_{i\omega}$	Speed integral gain	1000
N_p	Prediction horizon	4
N_c	Control interval	1
a_1	Laguerre factor for d-axis voltage	0.9
a_2	Laguerre factor for q-axis voltage	0.9
N_1	Number of Laguerre terms for d-axis voltage	4
N_2	Number of Laguerre terms for q-axis voltage	4
q_{i_d}	Weight of d-axis current error	10,000
q_{i_q}	Weight of q-axis current error	10,000
q_{mtpa}	Weight of MTPA trajectory error	10,000
r	Weight of voltage increment error	0.1

The MTPA tracking performance of Motor 2 under load variation is illustrated in Figure 3, which presents the dynamic current response trajectories of three control strategies: the proposed Laguerre-MTPA MPC, the traditional MTPA method, and standard MPC without MTPA integration. In this simulation, the reference speed of Motor 2 is set to 100 rad/s with an initial external load torque of 10 N · m. At $t = 0.3$ s, the load increases to 20 N · m, and at $t = 0.6$ s, it decreases back to 10 N · m. Figure 3 shows the current trajectories in the d-q plane, which are used to evaluate the tracking capability of each control method with respect to the theoretical MTPA trajectory calculated from Equation (9). It is observed that the traditional MTPA method exhibits noticeable current fluctuations and significant deviation from the optimal trajectory. In contrast, the standard MPC without MTPA exhibits a wide and scattered current distribution, lacking consistency

and alignment with the desired torque-per-ampere path. The proposed Laguerre-MTPA MPC strategy, however, demonstrates a highly concentrated current trajectory that closely follows the theoretical MTPA curve, indicating superior tracking accuracy and improved control consistency under dynamic loading conditions.

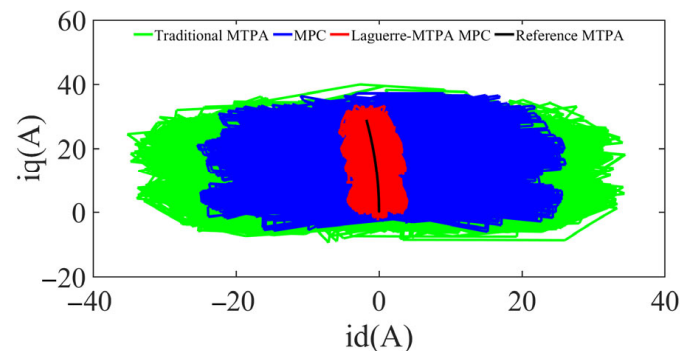


Figure 3. MTPA response of Motor 2 under varying load conditions.

In contrast to Figure 3, the MTPA tracking performance of Motor 1 under load variation is shown in Figure 4. To further verify the generality and robustness of the proposed Laguerre-MTPA MPC strategy across different types of PMSMs, Motor 1 is selected for comparative simulation. The load variation scenario is kept identical to that in Figures 3–5 for consistency. Unlike Motor 1, which features similar d-axis and q-axis inductances and thus a minimal contribution from reluctance torque, Motor 2 exhibits a significant difference between the two inductances (i.e., $L_d \neq L_q$), resulting in a pronounced magnetic saliency and a noticeably nonlinear MTPA trajectory. As shown in the d-q current distribution plot in Figure 4, the intrinsic motor characteristics clearly influence the performance of each control strategy. For Motor 2, although both the traditional MTPA control and the standard MPC show deviations from the ideal MTPA path, the overall current distribution remains relatively compact. However, for Motor 1 with its stronger reluctance torque characteristics, the traditional MTPA control results in greater deviations from the theoretical trajectory, while the current point cloud under standard MPC becomes highly scattered and unstable, severely compromising current utilization and torque output precision. In contrast, the proposed Laguerre-MTPA MPC algorithm demonstrates consistent tracking performance across both motor types. Even in the presence of dominant reluctance torque effects in Motor 1, the current trajectory remains closely aligned with the theoretical MTPA path. No significant current deviation or instability is observed, highlighting the superior adaptability and robustness of the proposed method under varying magnetic saliency conditions.

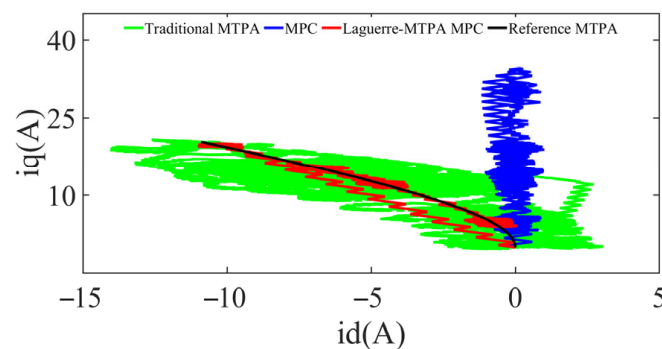


Figure 4. MTPA response of Motor 1 under varying load conditions.

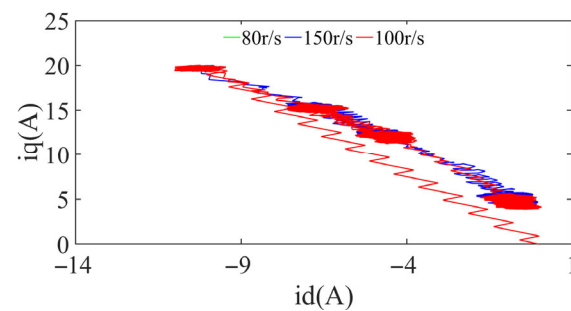


Figure 5. MTPA response of Motor 1 under varying speed conditions.

The MTPA tracking performance of Motor 1 under varying speeds is illustrated in Figure 5. This figure presents the current distribution trajectories of the proposed Laguerre-MTPA MPC strategy under different speed conditions, while keeping the load variation scenario identical to that described in Figure 3. Three representative operating points, 80 rad/s, 100 rad/s, and 150 rad/s, are selected for comparative simulation, and the corresponding current paths are plotted in the d-q plane. As shown in Figure 5, the current trajectories at all three speeds are well aligned with the theoretical MTPA curve and exhibit high compactness and consistency, indicating strong speed adaptability of the proposed controller. Specifically, as the speed increases, the spread of the current along the d-axis becomes slightly wider due to the increased torque demand and system dynamics. Nevertheless, the Laguerre-MTPA MPC controller consistently maintains accurate tracking of the MTPA trajectory without exhibiting notable deviation or oscillatory behavior. Notably, under the high-speed condition of 150 rad/s, where system dynamics are more pronounced, the controller still generates prompt and stable current commands that closely follow the optimal torque-per-ampere path. This reflects the effectiveness of Laguerre function-based parametrization in maintaining optimization stability even at elevated speeds. These results confirm that the proposed Laguerre-MTPA MPC strategy is not only effective under steady-state or low-speed conditions but also exhibits excellent trajectory tracking accuracy and robustness at high speeds, making it well-suited for wide-speed-range electric drive applications.

The electromagnetic torque performance under load variation is compared in Figure 6, which illustrates the response of three control strategies: the proposed Laguerre-MTPA MPC, the traditional MTPA-MPC, and the standard Laguerre-based MPC (LMPC). The test scenario mirrors the one used in previous sections; Motor 1 operates at a reference speed of 100 rad/s with an initial external load torque of 10 N·m. At $t = 0.3$ s, the load increases to 20 N·m, and at $t = 0.6$ s, it decreases back to 10 N·m. As observed in Figure 6, although all three control strategies are capable of responding to load variations, their performance differs significantly. The standard LMPC (without MTPA integration) exhibits a slower response speed, noticeable overshoot, and persistent steady-state oscillations. The traditional MTPA-MPC improves the response speed but shows slight oscillations at moments of abrupt load change. In contrast, the proposed Laguerre-MTPA MPC achieves superior dynamic performance with faster response, reduced overshoot, and smoother torque convergence during both disturbance events. For instance, in the load step-up scenario at $t = 0.3$ s, the Laguerre-MTPA MPC stabilizes the electromagnetic torque near the new setpoint within approximately 10 ms, about 30% faster than the traditional MTPA-MPC method. Similarly, during the load drop at $t = 0.6$ s, the proposed controller demonstrates lower response delay and quicker recovery to steady state, further confirming its effectiveness in handling abrupt load disturbances with high precision and robustness.

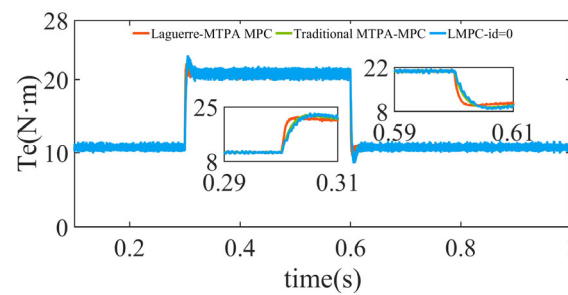


Figure 6. Electromagnetic torque response comparison under load changes.

Figure 7 presents the current magnitude response curves under load disturbance for different control strategies, aiming to evaluate their impact on current utilization efficiency. The load variation scenario is identical to that described in Figure 6. As shown in the figure, all three control methods exhibit significant changes in current magnitude during load transients, but their dynamic behaviors differ markedly. The standard LMPC maintains relatively high current magnitudes throughout the entire process, with sharp fluctuations during load transitions, indicating poor current utilization and lower efficiency. The traditional MTPA-MPC strategy shows improved performance in steady-state phases, demonstrating some degree of current optimization. However, it still suffers from abrupt current spikes and minor oscillations during load disturbances. In contrast, the proposed Laguerre-MTPA MPC consistently achieves lower current magnitude responses, with faster convergence and smaller fluctuations during transient periods. Notably, during the load increase phase at $t = 0.3$ s, the current under Laguerre-MTPA MPC rapidly rises to the new steady-state level and stabilizes quickly, demonstrating excellent dynamic current control capability. Moreover, in steady-state operation, the proposed method reduces the average current magnitude by approximately 12–18% compared to the other methods, shortens the transient settling time by around 25%, and exhibits smoother amplitude transitions. Figure 7 shows that the proposed Laguerre-MTPA MPC achieves a reduction of approximately 12–18% in the average current magnitude compared with the baselines. Since copper loss is given by $P_{Cu} = 3I^2R_s$ (with R_s listed in Table 2), this directly corresponds to a 12–18% reduction in copper loss and, thus, in winding thermal stress. The smoother current trajectories also imply fewer transient spikes, contributing to lower cumulative energy consumption over drive cycles. These improvements directly contribute to lower copper losses, higher system energy efficiency, and reduced thermal stress.

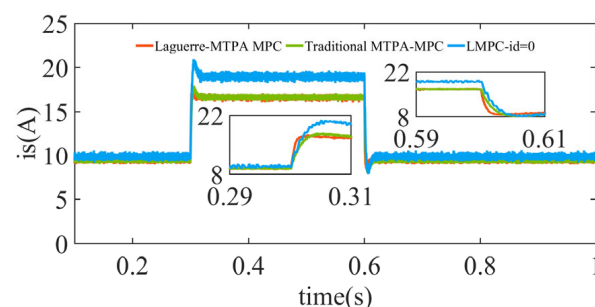


Figure 7. Current magnitude response comparison under load changes.

Figure 8 illustrates the speed response of the motor under the same load disturbance conditions, comparing the ability of each control strategy to reject disturbances and maintain speed stability. Although all controllers maintain generally stable speed outputs, varying degrees of speed fluctuations occur during sudden load changes. The standard LMPC strategy shows the largest deviations, with significant speed dips and

overshoots at both disturbance moments, along with the longest recovery time. The traditional MTPA-MPC demonstrates improved disturbance rejection but still exhibits notable speed deviations when subjected to abrupt load changes. In comparison, the proposed Laguerre-MTPA MPC controller achieves the smallest speed fluctuations and the fastest dynamic recovery during both disturbance events. As shown in the detailed insets, during the sudden load increase at $t = 0.3$ s, the proposed controller results in the smallest speed drop and restores the steady-state value within approximately 10 ms. Similarly, during the load decrease at $t = 0.6$ s, it produces the smallest overshoot and exhibits a smooth convergence trend, outperforming both benchmark methods. These results indicate that the proposed Laguerre-MTPA MPC not only improves electromagnetic torque control precision but also significantly enhances the system's disturbance rejection capability under variable operating conditions.

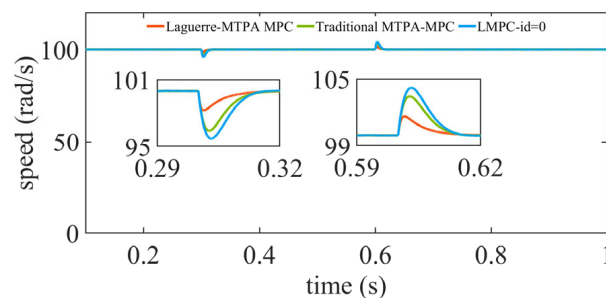


Figure 8. Speed tracking response comparison under load changes.

Figures 9 and 10 illustrate the responses of the d-axis and q-axis currents, respectively, under the load variation scenario described in Figure 6. These current trajectories are used to assess the ability of different control strategies to adapt to changing torque demands, particularly reflected in the behavior of the q-axis current, which is directly related to electromagnetic torque generation. As shown in Figure 10, the standard LMPC (configured with baseline parameters) exhibits the largest q-axis current fluctuations at both disturbance moments, accompanied by pronounced oscillations during the transient phases and poor overall stability. The traditional MTPA-MPC strategy offers some improvement in suppressing fluctuations but still shows brief overshoots during load transitions. In contrast, the proposed Laguerre-MTPA MPC demonstrates the most stable q-axis current response under both disturbance events. The current transitions are smooth, with minimal fluctuations, and the steady-state values accurately align with the torque levels corresponding to the external load changes. A closer examination of the detailed insets reveals that during the load increase at $t = 0.3$ s, the q-axis current under the proposed controller rises rapidly and settles smoothly, whereas the other strategies exhibit either delay or oscillatory behavior. Similarly, during the load decrease at $t = 0.6$ s, the Laguerre-MTPA MPC shows a faster downward response with the smallest overshoot.

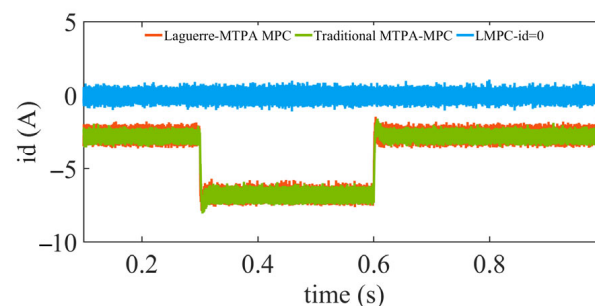


Figure 9. d-axis current response under load changes.

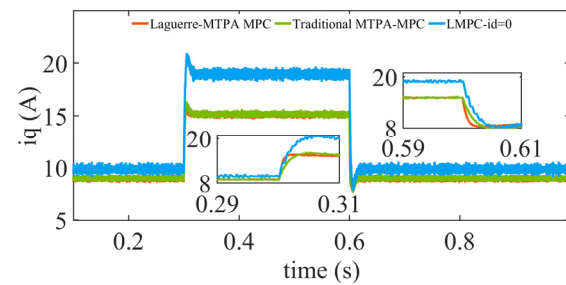


Figure 10. q-axis current response under load changes.

Overall, these results confirm that the proposed controller not only refines the current trajectory for torque generation but also significantly improves the system's dynamic responsiveness and steady-state precision under load disturbances. This highlights the superior electromagnetic control capability and dynamic consistency of the Laguerre–MTPA MPC strategy.

5. Conclusions

This paper presents a novel energy-efficient control strategy that integrates Laguerre function-based modeling with MTPA optimization to enhance the performance of PMSM drive systems in electric vehicles. By compressing the control input sequence using Laguerre orthogonal functions, the proposed method significantly reduces the dimensionality of the optimization problem, thereby improving the real-time computational efficiency and enabling practical online deployment of the MPC controller. In addition, an equivalent MTPA current trajectory is dynamically constructed to generate optimal current references that maximize torque output per unit current. This integration not only enhances the responsiveness of the controller but also improves the current utilization efficiency and aligns with energy-saving objectives. In this work, the proposed Laguerre–MTPA MPC framework has been shown to reduce the current magnitude, improve dynamic performance, and enhance robustness under parameter variations and load disturbances. These improvements translate directly into copper loss reduction and energy savings. In future work, we plan to extend the evaluation to full drive-cycle energy accounting (e.g., WLTP-like torque–speed profiles), including cumulative kWh calculation, inverter switching losses, and iron/core loss estimation. Such studies will further confirm the practical energy-saving potential of the proposed controller in realistic automotive applications.

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